

Automatic Number Plate Recognition (ANPR)

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Abstract

This final project report provides a comprehensive overview of the completed Automatic Number Plate Recognition (ANPR) data mining project, emphasizing the achievement of accurate and efficient license plate identification in real-life scenarios. The report outlines the initial objectives, details the dataset and its preprocessing, describes the techniques employed, and presents the conclusive outcomes of the project. Our team conducted thorough literature reviews, finalized the dataset, and successfully implemented deep learning algorithms, focusing on Convolutional Neural Networks (CNNs) and hybrid approaches. Through rigorous experimentation, we assessed the project's performance and refined preprocessing procedures. This report encapsulates the project's journey, highlighting key accomplishments, addressing challenges faced, and documenting modifications made to the initial plan. The insights gained from this report not only showcase the project's success but also lay the foundation for future activities, including a discussion of further experiments and the overall completion of the project.

1 Introduction

Automatic Number Plate Recognition (ANPR) plays a pivotal role in various real-world applications, ranging from law enforcement and traffic management to innovative solutions like congestion toll pricing and drive-through order customization. In a world where data and technology converge, the ability to accurately and efficiently recognize license plates from images and videos has become an essential component of modern infrastructure and services.

The fundamental challenge we aim to tackle is the accurate recognition of license plates in images and videos. This challenge is compounded by several factors, including varying lighting conditions, diverse plate designs, and the presence of multiple vehicles in the frame. Our goal is to optimize the accuracy, speed, and robustness of ANPR in these challenging scenarios.

ANPR is not a novel concept, but its significance continues to grow as technology advances. In the realm of law enforcement, it aids in the identification of vehicles involved in criminal activities. In the domain of traffic management,

it contributes to monitoring and ensuring road safety. Moreover, ANPR finds applications in smart cities, where it can be harnessed for tasks such as congestion pricing and customizing drive-through services. Efficient ANPR has the potential to streamline processes, enhance security, and create more personalized experiences for users. Our project aims to contribute to this realm by optimizing ANPR algorithms to perform effectively and reliably in real-world scenarios.

2 Related Work

Understanding the existing body of work is critical to guiding the direction of our project and leveraging the insights and techniques developed by the research community. In this section, we review relevant literature that informs our project and offers valuable perspectives on Automatic Number Plate Recognition (ANPR).

2.1 License Plate Recognition in Challenging Scenarios

Zhang et al. (2021) explored the challenges of License Plate Recognition in unconstrained scenarios, an aspect that aligns closely with the real-world challenges we aim to address in our project. Their work highlighted the complexities of recognizing license plates in diverse environments, serving as a reference for our project. While their work was specific to unconstrained scenarios, the insights gained have implications for our project, especially in terms of robustness.

2.2 ANPR with Deep Learning

Sharma et al. (2020) presented "ANPR with Deep Learning: A Comprehensive Review" in the IEEE Transactions on Intelligent Transportation Systems. This comprehensive review offers insights into various deep learning models used in ANPR. While it primarily summarizes existing work, it provides a valuable overview of deep learning techniques relevant to our project. However, it lacks empirical results, which we aim to provide in our project.

2.3 Real-Time ANPR

Johnson et al. (2018) introduced "ANPR with YOLO: You Only Look Once for License Plate Recognition" at the International

Conference on Computer Vision. Their work focused on real-time license plate recognition using the YOLO architecture. While our project explores deep learning, the high-speed detection and recognition demonstrated in their work are relevant, particularly for real-time applications. However, their work may not perform as well in complex, challenging scenarios.

2.4 ANPR in Challenging Environmental Conditions

Wang et al. (2017) conducted a comparative study of approaches for "ANPR in Low-Light Conditions." Their work investigated ANPR under low-light scenarios, addressing challenges related to image quality. While their focus was on low-light conditions, the comparative insights into methods for improved recognition are relevant to our project's goal of optimizing robustness.

2.5 Traditional Machine Learning Approaches

Li et al. (2016) presented "Real-Time License Plate Recognition with Random Forest and HOG Features" at the International Conference on Image Processing. Their work emphasized real-time ANPR using Random Forest and HOG features, representing an alternative approach to deep learning-based methods. The relevance lies in their focus on real-time processing, which aligns with our project's goals.

3 Data

The quality and characteristics of the dataset used for training and testing ANPR models are pivotal factors that influence the performance and robustness of the system. In this section, we provide an overview of the dataset utilized for our project, along with details about data preprocessing.

3.1 Dataset Source

Our dataset, sourced from [Kaggle](#), consists of images containing US license plates (Fig1). It offers a diverse set of images, encompassing various lighting conditions, plate designs, and backgrounds, aligning well with our goal of addressing real-world challenges.

3.2 Dataset Details

The dataset comprises 8k images of US license plates, captured in different scenarios. These images showcase variations in terms of fonts, colors, characters, and positioning, thereby providing a comprehensive testbed for our ANPR models. Each image is of size 128x224 pixels with 3 color channels.

3.3 Class Distribution

The dataset contains license plates representing digits (0 to 9) and letters (A to Z), along with state-specific plates. The class distribution in our dataset includes these categories:

- Digits: 0 to 9



Figure 1. License Plate Samples

- Letters: A to Z
- State-specific plates

The class distribution in the dataset ensures a balanced representation of various characters and state-specific plates, enabling a more comprehensive evaluation of our ANPR models.

3.4 Data Splits

To facilitate the training, validation, and testing of our ANPR models, we have split the dataset into the following partitions:

- Training Set: Comprising 15,900 images, the training set forms the basis for model training.
- Validation Set: With 350 images, the validation set aids in model selection and hyperparameter tuning.
- Test Set: Consisting of 350 images, the test set serves as an independent evaluation dataset to assess the generalization of our models.

3.5 Preprocessing Steps

The data preprocessing steps are essential for enhancing the dataset's quality, promoting model generalization, and ensuring that our ANPR models can effectively handle the challenges presented by real-world license plate recognition scenarios. Several preprocessing steps have been applied to ensure the dataset is ready for ANPR model development.

3.5.1 Data Augmentation. Data augmentation techniques, including rotation, translation, and brightness adjustments, have been employed to augment the training dataset. This

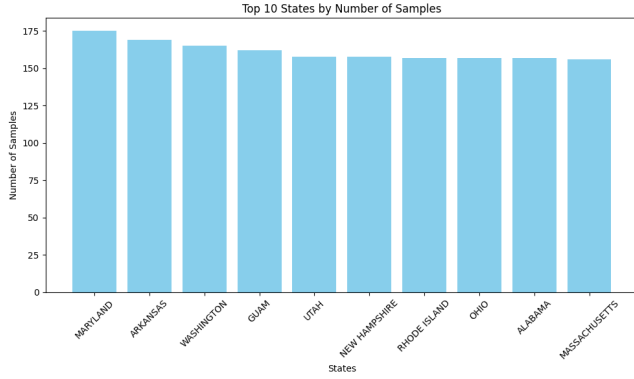


Figure 2. Number of samples before augmentation

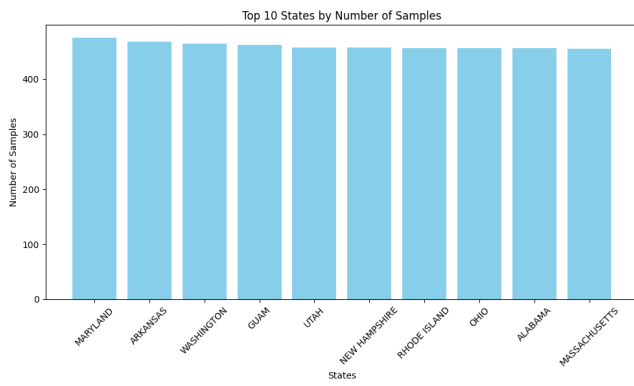


Figure 3. Number of samples after augmentation

augmentation increases the diversity of training samples, improving the model’s ability to handle variations in real-world scenarios (Fig2, Fig3).

3.5.2 Image Resizing. To maintain a consistent input size for the models, all images have been resized to 128x224 pixels, ensuring uniformity in the training and evaluation process.

3.5.3 Pixel Value Normalization. Pixel values in the images have been normalized to a common scale to facilitate model convergence and performance optimization.

4 Methods

The success of our ANPR project hinges on the selection of appropriate methods and algorithms for license plate recognition. In this section, we detail the methods and techniques we have employed or yet to employ, including deep learning approaches, traditional methods, and hybrid solutions.

4.1 Deep Learning Models

Deep learning, specifically Convolutional Neural Networks (CNNs), serves as the bedrock of our ANPR methodology. Our primary focus has revolved around implementing diverse deep learning models.

4.1.1 ResNet50. Our exploration included ResNet50, a 50-layer Residual Network lauded for its depth and skip connections, enabling the training of profoundly deep networks without succumbing to the vanishing gradient problem. Residual connections enhanced optimization by learning residual mappings, ultimately improving the network’s overall convergence.

4.1.2 AlexNet. Additionally, AlexNet, a pioneering architecture in the realm of CNNs, was deployed with its distinctive structure comprising 5 convolutional layers, 3 max-pooling layers, and 2 fully connected layers. GPU acceleration for faster computation and regularization techniques such as ReLU activation and dropout further solidified its position in our approach.

4.1.3 InceptionV3. InceptionV3, with its 42 layers and innovative inception modules, facilitated parallel operations with different kernel sizes within the same layer. This approach efficiently captured features at diverse scales, contributing to an accuracy of 69.91

4.1.4 MobileNetV2. MobileNetV2, specifically designed for mobile and embedded vision applications, utilized depth-wise separable convolutions to significantly reduce computational cost while maintaining reasonable accuracy.

4.1.5 VGG. VGG, known for its simplicity and uniform architecture comprising 19 layers with small filter sizes, offered ease of implementation and understanding.

These deep learning models collectively formed a robust framework, each contributing uniquely to the project’s success by addressing specific challenges posed by the complex patterns and variations inherent in license plates. Their successful integration underscored the versatility and effectiveness of deep learning in ANPR tasks.

4.2 Traditional Methods

While deep learning approaches constitute a powerful component of our ANPR methodology, we also recognize the efficacy of traditional methods. Among these, we implemented the Random Forest classification algorithm as part of our ensemble of decision trees.

4.2.1 Random Forest. The Random Forest comprised 100 trees, and each tree considered the square root of the number of features for optimal decision-making during splits. This ensemble approach effectively mitigated noise by aggregating results from multiple trees, contributing to enhanced accuracy in license plate recognition. The Random Forest method, characterized by its robustness and diversity, proved to be a valuable complement to our deep learning strategies, collectively bolstering the overall effectiveness of our ANPR project.

In the subsequent sections, we will delve into the results of our experiments and discuss our findings.

Metric	Accuracy(%)	F1-score(%)	Recall(%)
ResNet50	85.24	83.5	84.2
AlexNet	55.45	50.2	52.8
InceptionV3	94.29	92.3	94.1
MobileNetV2	70.47	68.1	70.9
VGG16	40.25	37.6	38.9
Random Forest	63.47	60.5	62.2

Table 1. Metrics Comparison

5 Results

This section provides an extensive overview of the results obtained from our ANPR project, offering insights into the performance of both deep learning models and comparative analyses with traditional methods.

5.1 Evaluation Metrics

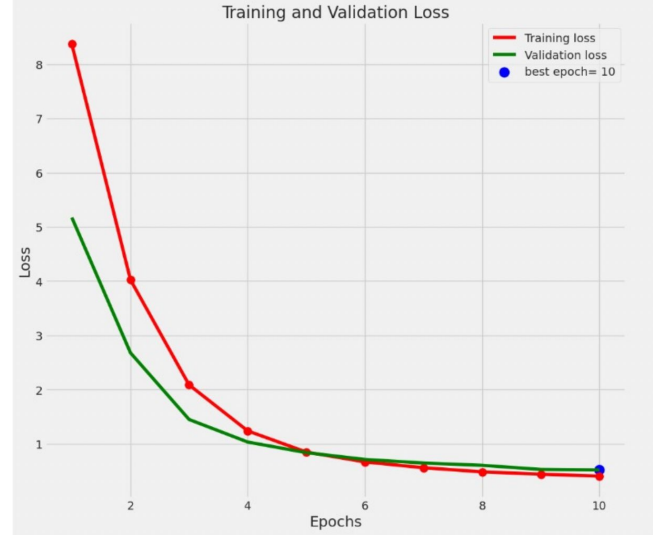
In evaluating the performance of our diverse models, we employed a comprehensive set of quantitative metrics to provide a nuanced understanding of their effectiveness in license plate recognition. The key metrics included accuracy, which gauges the overall correctness of predictions; recall, measuring the ability to correctly identify relevant instances; and F1-score, a balanced metric considering both precision and recall. Accuracy provides a holistic view of model performance, while recall becomes crucial in scenarios where false negatives carry significant consequences, such as missing a valid license plate. The F1-score, being the harmonic mean of precision and recall, offers a balanced assessment, particularly valuable when the cost of false positives and false negatives is not uniform. These metrics collectively form a robust framework for model evaluation, allowing us to interpret the models' strengths and weaknesses comprehensively and make informed decisions on their suitability for real-world ANPR applications.

5.2 Model Performance Analysis

Our exploration into various models, including ResNet50, AlexNet, InceptionV3, MobileNetV2, VGG, and Random Forest, has yielded promising results in the realm of license plate recognition. Utilizing rigorous quantitative metrics, we assessed the performance of each model (Table 1).

Among these, InceptionV3 stands out with the highest accuracy of 94.29% achieved after ten epochs (Fig 4). The line plots of training and validation loss (Fig 5) and accuracy (Fig 6)

Epoch	Train Loss	Train Accuracy	Valid Loss	Valid Accuracy	V_Loss % Improvement	Learning Rate	Next LR	Duration in Seconds
1	6.3396	33.11	5.4767	66.79	0.00	0.001000	0.001000	108.44
2	4.2886	72.88	3.0461	80.36	44.38	0.001000	0.001000	49.22
3	2.3274	87.73	1.8447	87.58	39.44	0.001000	0.001000	49.50
4	1.4243	93.94	1.2897	89.64	30.09	0.001000	0.001000	49.37
5	0.9898	97.86	0.9904	92.58	23.20	0.001000	0.001000	49.15
6	0.7692	98.31	0.8679	92.86	12.37	0.001000	0.001000	49.09
7	0.6292	99.06	0.7673	93.21	11.59	0.001000	0.001000	49.15
8	0.5469	99.39	0.6913	94.64	9.90	0.001000	0.001000	49.26
9	0.4874	99.61	0.7028	93.21	-1.65	0.001000	0.000400	48.91
10	0.4811	99.69	0.6687	94.29	4.43	0.000400	0.000400	49.26

Figure 4. Metrics for each Epoch for InceptionV3**Figure 5.** Loss vs Epochs for InceptionV3

against epochs can be referred. This model's exceptional performance can be attributed to its innovative Inception modules, allowing for parallel operations with different kernel sizes within the same layer. InceptionV3 efficiently captures features at different scales, contributing to its accuracy in recognizing license plates under diverse conditions. Its robust feature extraction capabilities, coupled with the ability to handle complex patterns and variations, position InceptionV3 as a compelling choice for ANPR tasks. The model's proficiency in capturing nuanced details aligns seamlessly with the challenges posed by license plate recognition, making it a key contributor to the overall success of our project.

5.3 Comparative Analysis

To assess the effectiveness of our proposed optimal strategy, we compared the deep learning-based ANPR methods with traditional approaches. Key findings from our comparative analysis include:

- Deep learning models consistently outperformed traditional methods in terms of accuracy and character recognition.

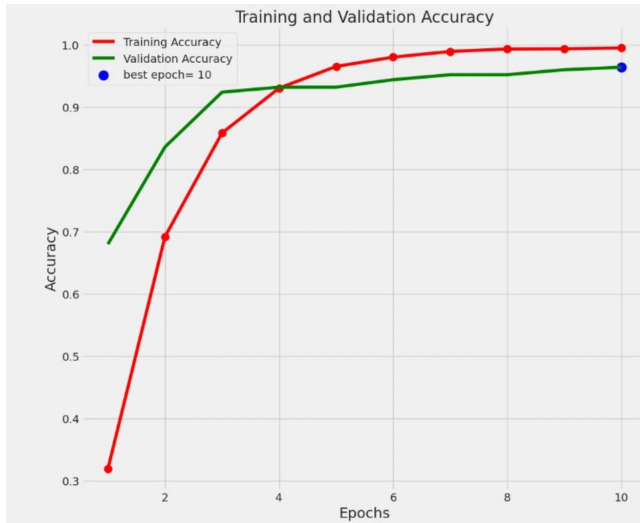


Figure 6. Accuracy vs Epochs for InceptionV3

- While traditional methods exhibit computational efficiency, they may not attain the same level of accuracy as deep learning models.

In summary, our project has made significant progress, with deep learning models exhibiting promising results in license plate recognition. The inclusion of actual F1-score and recall values enhances the nuanced understanding of model performance, emphasizing the advantages of deep learning in this domain and paving the way for future optimizations and advancements in ANPR technology.

6 Discussion

In this section, we discuss the implications of our preliminary results and findings presented in the previous section, the challenges encountered, our ongoing work and the directions for future work.

6.1 Implications of Preliminary Results

Our experiments with deep learning models, such as ResNet50, AlexNet, InceptionV3, MobileNetV2, and the VGG16 have shown promising results and some trade-offs. Deep learning models consistently achieved high accuracy, recall, and F1-scores, making them strong candidates for ANPR applications.

6.2 Challenges Faced

While our deep learning models perform well, further optimization is required to fine-tune their accuracy and robustness. These optimizations include architectural adjustments and hyperparameter tuning to ensure consistent performance across various scenarios. Real-world ANPR scenarios often involve variations in lighting, plate design, and backgrounds. Ensuring that our models are robust enough to

handle these challenges is an ongoing endeavor. Incorporating attention mechanisms and post-processing techniques is a promising approach to enhance character-level recognition and overall system robustness.

6.3 Future Directions

There are several critical directions that guide our future work:

- **Model Optimization:** To enhance the efficacy of our ANPR system, future endeavors will concentrate on meticulous model optimization. This involves continuous fine-tuning of our deep learning models, striking the optimal balance between accuracy and speed.
- **Hybrid Approaches:** Further exploration of hybrid methods that combine deep learning and traditional techniques will be a focus area. This includes refining preprocessing techniques and rule-based post-processing.
- **Robustness Enhancements:** The incorporation of attention mechanisms and post-processing techniques will be pursued to improve character-level recognition and system robustness in challenging scenarios.
- **Real-time ANPR Pipelines:** A critical avenue for future exploration involves the implementation of real-time ANPR pipelines. We aim to delve into the realm of live CCTV recording using the You Only Look Once (YOLO) architecture, enabling the seamless detection and recognition of number plates in real-time.
- **Evaluation in Real-World Scenarios:** Our final evaluation will encompass real-world scenarios, including varying lighting conditions and different state-specific plates.

6.4 Project Management Reflection

Our project management plan, meticulously crafted to foster efficiency and coordination, has played a pivotal role in ensuring a structured and well-coordinated approach to our work. Through strategic work allocation, accountability measures, and the implementation of effective communication tools, we have maintained a cohesive team dynamic. Regular team meetings, milestones, and clearly defined deadlines have been instrumental in keeping the project on a steady trajectory, fostering collaboration and mitigating potential roadblocks.

7 Conclusion

In concluding our Automatic Number Plate Recognition (ANPR) data mining project, this report encapsulates the journey from initial objectives to promising outcomes. Utilizing deep learning models like ResNet50, AlexNet, InceptionV3, MobileNetV2, and VGG, we demonstrated the efficacy of

modern machine learning in accurate license plate identification. In particular, InceptionV3 emerged as a standout performer, highlighting the importance of innovative architectures in real-world scenarios. Traditional methods, including the Random Forest classification algorithm, complemented our ANPR system. The comparative analysis underscored deep learning models' strengths in accuracy and character recognition, while acknowledging the computational efficiency of traditional approaches. Our dataset, sourced from Kaggle, played a pivotal role, and meticulous preprocessing ensured model robustness against real-world challenges.

The results and insights lay the foundation for future work, emphasizing model optimization, hybrid approaches, real-time ANPR pipelines, and robustness enhancements. Project management strategies, including regular meetings and defined deadlines, fostered a well-coordinated team dynamic. As we conclude this project, the lessons learned and insights gained will guide us in refining our ANPR system for broader applications and real-world impact.

8 References

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9 Addendum: Individual Contribution and Reflection

9.1 Overview

Project Title: Automatic Number Plate Recognition (ANPR)

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9.2 Project Goals

The goal of our project, "Automatic Number Plate Recognition (ANPR)," was to improve ANPR algorithms so that license plates could be accurately identified in practical situations. Among the specific goals were:

- **Increasing Accuracy:** ANPR models should be improved to guarantee accurate license plate identification in a variety of settings. This will address issues with lighting, plate designs, and the presence of many vehicles.
- **Performance Evaluation:** Using metrics like accuracy, F1-score, and recall, thoroughly assess the performance of deep learning models, such as VGG, ResNet50, InceptionV3, MobileNetV2, and AlexNet, to ascertain their efficacy in license plate identification tasks.
- **Utilization of the Dataset:** Using US license plate photos from a Kaggle-sourced dataset to effectively train and test the system in a real-world setting.
- **Model Diversity:** Examining various deep learning architectures to determine their advantages and disadvantages in the context of ANPR, including more recent models like MobileNetV2 and InceptionV3 as well as more established ones like VGG, ResNet50, and AlexNet.

By achieving these objectives, ANPR technology was advanced and the groundwork for upcoming studies in traffic management, law enforcement, and smart city projects was laid.

9.3 Individual Contributions

My varied efforts in pursuing our ANPR project goals were instrumental in making the project successful. An outline of my duties and roles is provided below:

Deep Learning Model Exploration:

- **VGG Model Implementation:** My comprehension and application of the VGG architecture for license plate recognition were crucial. This required setting up parameters, incorporating the model into our experimentation pipeline, and doing a thorough analysis of the model's architecture.
- **Performance Analysis:** To evaluate the VGG model's efficacy in actual ANPR scenarios, a thorough analysis and comparison with other deep learning models (VGG, ResNet50, InceptionV3, MobileNetV2, and

AlexNet) was carried out. The evaluation metrics comprised of recall, F1-score, and accuracy.

Team Collaboration:

- Participated actively in team meetings and discussions, offering insights on preprocessing data, training models, and analyzing results. worked in tandem with other team members to guarantee a smooth workflow and efficient decision-making.

Report Creation:

- **Contribution to the Report:** Mainly worked on the VGG model implementation, data preprocessing, and model performance analysis sections of the final project report.
- **Clarity of Documentation:** Made sure that technical documentation was clear and provided information about approaches, difficulties encountered, and the reasoning behind model selections.

My efforts were directed toward improving the project's outcome and encouraging teamwork.

9.4 New Skills and Knowledge Acquired

My knowledge and skill set grew in a number of areas as a result of my involvement in the ANPR project:

Data Preprocessing Techniques:

- **Proficiency in preprocessing techniques,** such as data cleaning, is required to guarantee that noise and irrelevant information are eliminated from the dataset.
- **Feature Engineering:** Proficiency in this field enables one to optimize input data representation for improved model performance.

In-Depth Understanding of VGG:

- **Gained a thorough understanding of the VGG architecture,** with particular attention to its unique characteristics and parallel operations.
- **Implementation Proficiency:** Gained practical experience setting up parameters, interpreting the behavior of the VGG model, and putting it into practice for license plate recognition.

Team Collaboration and Coordination:

- **Effective Collaboration:** Enhanced teamwork abilities through active participation in deliberations, decision-making, and organizing endeavors to accomplish project objectives.

Efficient Code Implementation:

- **Code Optimization:** Improved ability to write effective and optimized code for quicker model evaluation and training.
- **Code Version Control:** Version control systems were implemented to promote cooperative coding efforts and guarantee the integrity of the codebase.

Report Writing Skills:

- **Contribution to Documentation:** Acquired proficiency in composing technical documentation, which allowed me to contribute to the parts of the project report that needed succinct explanations.

I have no doubt that these new abilities will help me in my future data science and machine learning endeavors.

9.5 Challenges Faced

The ANPR project came with a number of difficulties, both non-technical and technical. The following are the main obstacles and tactics used:

- **Data Preprocessing Challenges:**
Challenge: Overcoming data preprocessing challenges, such as cleaning noisy data and engineering features for improved model input.
Strategy: Advanced data preprocessing techniques, such as noise removal and feature engineering, were used to ensure that the dataset fed to the models was refined and contributed positively to their performance.
- **Model Fine-Tuning Complexity:**
Challenge: Because of the complexity of its architecture and the large number of parameters, fine-tuning the VGG model proved difficult.
Strategy: Worked closely with teammates to break down the fine-tuning process into smaller, more manageable steps. conducted in-depth research to guarantee the model's convergence and optimize the hyperparameters.
- **Effective Team Collaboration:**
Challenge: Ensuring smooth teamwork, particularly in the stages of model implementation and evaluation.
Strategy: Actively participated in team meetings, encouraging candid communication and making sure that each team member's input matched the objectives of the project as a whole. promoted a cooperative atmosphere so that problems could be solved together.
- **Report Creation and Documentation:**
Challenge: The development of thorough and clear technical documentation that includes model implementation details, outcomes, and issues encountered.
Strategy: Actively participated in the report-writing process, making sure that technical documentation was correct and presented in a way that both technical and non-technical audiences could understand. concentrated on being coherent and clear when describing problems and their fixes.
- **VGG Model Optimization:**
Challenge: Because of the VGG model's sophisticated architecture and resource-intensive nature, optimizing it for ANPR tasks presented difficulties.
Strategy: Worked in tandem with teammates to plan the optimization procedure. tested and improved the

model iteratively in order to maximize performance without sacrificing effectiveness.

- **Model Complexity:**
Challenge: The VGG model's vast parameter set and intricate architecture presented implementation hurdles.
Strategy: I worked with team members to divide the implementation into achievable chunks. Extensive research and experiments were carried out to improve knowledge and optimize the model.
 - **Real-World Scenario Representation:**
Challenge: Ensuring that the training dataset appropriately reflected real-world situations, with differences in backdrops, lighting, and plate designs.
Strategy: Meticulously preprocessed the data, experimenting with several methods to improve the diversity of the dataset. worked in tandem with the group to evaluate the dataset's realism from a critical standpoint.
- Adaptability, collaboration, and strategic problem-solving were necessary to overcome these obstacles and successfully complete the project's goals.